# ROBERT SIMON FONG

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## RESEARCH INTERESTS

- Differential Geometry, Information Geometry: theory and applications in Manifold Optimization
- Symplectic Geometry and Representational Learning
- Machine Learning and Dynamical Systems: Reservoir Computing
- Black-box Optimization: Bayesian Optimization, Zeroth-order Optimization

## EDUCATION

**University of Birmingham** *Sept. 2015 - July 2020*
*Ph.D., Computer Science*
Thesis: Stochastic Optimization on Riemannian Manifolds
Supervisors: Prof. Peter Tiňo, Prof. Joshua Knowles

**University of Waterloo** *Sept. 2013 - Aug. 2014*
*Master of Mathematics, Computational Mathematics*
Thesis: Optimization with Funcation Values Only
Supervisor: Prof. Thomas Coleman

**University of Waterloo** *Sept. 2007 - Apr. 2012*
*Bachelor of Mathematics, Pure Mathematics (Honours)*

## ACADEMIC POSITIONS

**Post-Doctoral Research Fellow (特任研究員)**, University of Tokyo *May. 2025 - present*

- **Research:** Symplectic Reservoir Computing
- Funded the International Research Center for Neurointelligence (WPI-IRCN) at The University of Tokyo Institutes for Advanced Study

**Research Fellow**, University of Birmingham *Feb. 2021 - present*

- **Research:** Reservoir Computing: Universality and applications on time series forecasting.
- **Teaching:** Mathematical Foundations of Artificial Intelligence and Machine Learning (MSc) [06 32250], Algorithms for Data Science (MSc) [06 32258]
- Funded by Alan Turing Institute; Prof. Peter Tiňo's Alan Turing Institute Fellowship
  *Machine Learning in the Space of State-Space Dynamic Models*

**Ph.D. Candidate**, University of Birmingham *Sept. 2015 - Jul. 2020*

- **Research** Manifold Optimization using Differential Geometry, Information Geometry and Simplicial Geometry.
- **Teaching**: MSc/ICY Software Workshop (Java) (1,2), Mathematical Foundations of Computer Science

**Master Student**, University of Waterloo *Sept. 2013 - Aug. 2014*

- **Research:** Derivative-free optimization: Simulated Annealing and Derivative-Free Zeroth-Order surrogate directional search methods. (with Prof. Thomas Coleman)

- **Research:** Computational Algebra: Solving multivariate polynomials using diagonal subgroup of linear group action. (with Prof. George Labahn)
- **Teaching:** Math 106 Linear Algebra for Arts (Co-organized).

**Undergraduate Research Assistant**, University of Waterloo *Jan. 2012 - Apr. 2012*

- Integer Programming and applications on Quary Crane Scheduling problem (special case of Traveling Salesman problem)

**Undergraduate Research Assistant**, Penn State University *Jan. 2009 - Mar. 2009*

- Optimization of Gravitational Wave Detectors in Laser Interferometer Gravitational-Wave Observatory(LIGO).

## INDUSTRY POSITIONS

**Senior Researcher**, Theory Lab, Huawei *Aug. 2022 - Mar. 2024*

- Quantum-Inspired Optimization – theory of Hamiltonian-based solver and its applications:
  - Boolean Satisfiability Problem, Electronic Design Automation (EDA), Hardware Verification, Graph partitioning, and MIMO Decoding.
- Reservoir Computing-based Time Series Forecasting
- Hardware Placement and Routing modelling in Integrated Circuits

**Researcher**, Noah's Ark Lab, Huawei *Aug. 2020 - Aug. 2021*

- Theory of Deep Neural Nets: convexity, modelling, and optimization
- Application and implementation of Deep Neural Nets and Bayesian Optimization:
  - 5G+ High Frequency Surface antenna design and Integrated Circuit Partition.

**Research Analyst**, Cayuga Research Associates *Jan. 2013 - Aug. 2015*

- Optimization and Simulation of Container Port control system.

**Consultant**, Hong Kong International Terminals Limited *Apr. 2012 - Jan. 2013*

- Integer Programming Modelling and Abstraction of ISO container placement

## BOOKS AND MONOGRAPHS

- **Robert Simon Fong** and Peter Tiňo. *Population-Based Optimization on Riemannian Manifolds*, volume 1046 of *Studies in Computational Intelligence (SCI)*. Springer, 2022. ISBN: 978-3-031-04292-8 (eBook ISBN: 978-3-031-04293-5)

## SELECTED JOURNAL ARTICLES

- **Robert Simon Fong**, Gouhei Tanaka, and Kazuyuki Aihara. Symplectic reservoir representation of legendre dynamics, 2025
- **Robert Simon Fong**, Boyu Li, and Peter Tiňo. Linear simple cycle reservoirs at the edge of stability perform fourier decomposition of the input driving signals. *Chaos*, 35(4):043109, 2025
- **Robert Simon Fong**, Yanming Song, and Alexander Yosifov. Efficient digital quadratic unconstrained binary optimization solvers for sat problems. *New Journal of Physics*, 27(1):013027, Jan 2025
- Boyu Li, **Robert Simon Fong**, and Peter Tiňo. Simple Cycle Reservoirs are Universal. *Journal of Machine Learning Research*, 25(158):1–28, 2024

- Robert Simon Fong. *Stochastic optimization on Riemannian manifolds.* PhD thesis, University of Birmingham, 2020

## SELECTED CONFERENCE PAPERS(REFERRED)

- Ziqiang Li, **Robert Simon Fong**, Kantaro Fujiwara, Kazuyuki Aihara, and Gouhei Tanaka. Structuring Multiple Simple Cycle Reservoirs with Particle Swarm Optimization. In *2025 International Joint Conference on Neural Networks (IJCNN)*, pages 1–9, 2025
- **Robert Simon Fong**, Boyu Li, and Peter Tino. Universality of real minimal complexity reservoir. *Proceedings of the AAAI Conference on Artificial Intelligence*, 39(16):16622–16629, 2025
- Peter Tiňo, **Robert Simon Fong**, and Roberto Fabio Leonarduzzi. Predictive Modeling in the Reservoir Kernel Motif Space. In *2024 International Joint Conference on Neural Networks (IJCNN)*, pages 1–8, 2024
- **Robert Simon Fong** and Peter Tiňo. Extended stochastic derivative-free optimization on riemannian manifolds. In *Proceedings of the Genetic and Evolutionary Computation Conference Companion*, GECCO '19, pages 257–258, New York, NY, USA, 2019. ACM
- **Simon Fong** and Peter Tiňo. Induced dualistic geometry of finitely parametrized probability densities on manifolds, 2018
- David Tsang, Andrew Lundgren, Ruxandra Bondarescu, **Simon Fong**, and Mihai Bondarescu. Optimizing finite mirrors for advanced gravitational wave detectors. In *APS April Meeting Abstracts*, 2009

## AWARDS AND SCHOLARSHIPS

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| **IRCN Retreat Brainstorming Award FY2025**, IRCN, The University of Tokyo | *2025* |
| **Honorary Research Fellowship**, University of Birmingham | *2021 - 2026* |
| **President's Scholarship**, University of Waterloo | *2008* |
| **AP Scholar with Distinction** , The College Board | *2007* |
| **Second Honour** , Third Pan Pearl Delta plus Chinese Elite Schools Physics Olympiad | *2007* |
| **Third Honour**, Hong Kong Physics Olympiad | *2007* |
| **Third Honour**, Hong Kong Physics Olympiad | *2006* |

## INVITED TALKS

**IRCN, The University of Tokyo**, Tokyo, Japan — *15. Apr. 2026*
*Symplectic Reservoir Representation of Legendre Dynamics*

**IRCN, The University of Tokyo**, Tokyo, Japan — *9. Jul. 2025*
*Universality of Simple Cycle Reservoirs*

**Drexel University**, Philadelphia, PA, USA — *28. Feb. 2025*
*Universality of Simple Cycle Reservoirs and Dilation Theory*

**With Prof. Boyu Li**
**Southern University of Science and Technology**, Shenzhen, China — *23. Aug. 2024*
*Universality of Simple Cycle Reservoirs*

**Huawei Noah's Ark Lab**, Hong Kong, China — *15. Mar. 2020*
*Information Geometry of Variational Autoencoder*

## JP NATIONAL CONFERENCE TALKS

**JST 若手数学者交流会（第 7 回)**, Tokyo, Japan — *21. Mar. 2026*
*Symplectic Reservoir Representation of Legendre Dynamics*

**Poster + Talk**

**第 3 回バイオ数理連携会議**, Tokyo, Japan *04. Dec. 2025*

*Simple Cycle Reservoirs: Universality and Applications*

**Poster**

**ICNSM2025**, Tokyo, Japan *19. Nov. 2025*

*Harmonics at the Edge of Stability*

**Poster**

**IRCN Retreat 2025**, Odawara, Japan *23. Oct. 2025*

*Simple Cycle Reservoirs: Universality in real numbers*

**Poster + Talk**

**IRCN Retreat 2025**, Odawara, Japan *23. Oct. 2025*

*A Particle Swarm Optimization Approach to Structuring Multiple Simple Cycle Reservoirs*

**Poster + Talk, with Z. Li, K. Fujiwara, K. Aihara, and G. Tanaka**

**IRCN WPI site visit 2025**, Tokyo, Japan *16. Jul. 2025*

*Harmonics at the Edge of Stability*

**Poster + Talk**

**IRCN WPI site visit 2025**, Tokyo, Japan *16. Jul. 2025*

*A Particle Swarm Optimization Approach to Structuring Multiple Simple Cycle Reservoirs*

**Poster + Talk, with Z. Li, K. Fujiwara, K. Aihara, and G. Tanaka**

## RESEARCH GRANTS

**JST/MS2: 第 3 回バイオ数理連携会議連携追加研究費**, **JPY 15,287,000** *2026-2028*

**Co-Investigator with 陳紫瑜 (片桐 PM・藤生克仁 PI), Lu Yurun (合原 PM・合原一幸 PI)**

*Ultra-early Prediction and Prevention of Heart Failure Using Multi-organ Dynamic Network Biomarkers and Sensory Neuromodulation*

**AI Incubator Startup Grant**, **JPY 1,000,000** *2026-2027*

**Principal Investigator**

*Symplectic Reservoir Learning for Regime Change Detection in Pre-disease Dynamics*

**IRCN Retreat Brainstorming Awards FY2025**, **JPY 800,000** *2025-2026*

**Principal Investigator**, 3 partners

*Multi-Reservoir Approach to Modular Invariance and Conservation in Biological Networks*

**AI Incubator Startup Grant**, **JPY 1,000,000** *2025-2026*

**Principal Investigator**

*Hamiltonian Representation in Reservoir Computing*

## REVIEW SERVICES

- Journal of Machine Learning Research (JMLR)
- SIAM Journal on Optimization
- ACM/SIGEVO Conference on Foundations of Genetic Algorithms (FOGA)